

Evidence-Bound Curation: Auditable Membership Decisions for Language-Model Pretraining Data

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Abstract—Large-scale data-curation pipelines combine heuristic filters, similarity retrieval, and learned quality models, but often leave these signals to directly determine corpus membership. We introduce evidence-bound curation, an auditable framework for language-model training data that separates evidence production from the authority to remove data. Validity handles closed input failures; Redundancy requires an explicit relation and a surviving representative; Quality positively selects data under declared evidence; and Coverage may veto removals that extinguish required support but cannot initiate deletion. Every membership transition is content-addressed and reason-coded, while downstream evaluation remains outside the curation runtime.

We instantiate the framework on a 6.98M-token Python corpus and construct Normal and Hard profiles under fixed policies. Hard retains 72.05% of the Raw token stream. Across three training seeds and six code benchmarks, Hard reaches a 20.80 primary reasoning macro compared with 20.86 for Raw, although benchmark-level changes are mixed. These results demonstrate auditable corpus compression with near-retained reasoning performance in the evaluated Code setting. The contribution is a domain-neutral membership-authority framework; broader downstream effectiveness requires independent validation in additional domains.

Index Terms—pretraining data, dataset curation, data quality, audibility, code language models

I. INTRODUCTION

Pretraining-data curation determines not only how much data a model sees, but which evidence remains available to learning. Large-corpus pipelines routinely remove malformed content, apply heuristic or learned quality filters, and deduplicate at exact, fuzzy, or semantic levels [1], [3]–[5]. Production tools such as NeMo Curator and Data-Juicer make these operations configurable and scalable [8], [10]. NeMo’s custom-filter interface usefully separates document scoring from the keep decision [11], but the recipe author still determines how heterogeneous decisions compose. That engineering progress exposes a narrower unresolved question: *when is a signal sufficient to change corpus membership?*

At large-corpus scale this is a data-systems problem, not merely a scoring problem: individually plausible operator

decisions can compound into millions of irreversible membership changes whose provenance is difficult to recover after materialization.

The question matters because the signals are not interchangeable. An absent payload is an input-contract failure. A MinHash collision is retrieval evidence, not proof that two programs are equivalent. A classifier abstention is not a negative quality judgment. Even when individual exclusions are locally defensible, their union can eliminate the only remaining member of a support group. A scalar score or ordered list of filters can conceal these differences behind one keep/drop bit.

Consider two units returned by a MinHash lookup. A similarity-first pipeline may treat retrieval as permission to discard one unit. Evidence-bound curation instead treats the pair as a candidate relation: removal is authorized only if the active policy verifies an allowed relation, records a surviving representative, and passes the post-composition Coverage recheck. The lookup is therefore useful without silently acquiring deletion authority. This pattern extends to learned Quality evidence and semantic support graphs.

We address the problem by separating *evidence* from *decision authority*. Evidence producers may describe a unit, retrieve comparisons, or estimate bounded properties, but only a versioned policy may perform a typed membership transition. Validity, Redundancy, Quality, and Coverage receive different authority and do not compensate for one another. Coverage is not a diversity reward or target-mixture optimizer: it is a final veto against unexplained support extinction. Downstream evaluation is outside the runtime and begins only after corpus identities are frozen.

Code is a useful stress domain for this separation. Whitespace-only variants may be equivalent, while a one-token operator, identifier, API signature, or negation change may alter behavior. Short fragments may be useful despite lacking execution context. Repository structure also produces long files, templates, generated artifacts, and source-correlated chunk distributions. These characteristics make aggressive similarity or scalar-quality shortcuts easy to state and difficult

to defend.

We evaluate the framework along two complementary dimensions. Operational validation tests whether typed authority prevents unsupported exclusions and preserves auditable corpus identities after policy composition. Downstream evaluation measures the compression–capability trade-off when the resulting profiles are used under their natural token budgets.

This paper makes three contributions:

- a typed membership-authority model with machine-checkable invariants for trace completeness, surviving representatives, zero-survivor support, forbidden inputs, and identity-bound replay;
- a concrete A–B–C pipeline combining candidate-only MinHash retrieval, survivor-linked redundancy witnesses, an abstention-aware vector Quality gate, and veto-only Coverage rematerialization; and
- a content-addressed implementation and frozen natural-budget evaluation protocol that exposes authority counterfactuals while preventing target size or downstream utility from entering curation.

The contribution is an executable decision contract that can sit above modern curation systems providing distributed filtering and deduplication. Its roles and transition types do not encode Code: evidence producers and versioned policy artifacts carry domain-specific assumptions. The framework is therefore language-model general at the authority layer, while the downstream evidence in this paper qualifies its Code instantiation only. Throughput improvements, new retrieval algorithms, and production readiness remain outside this claim.

II. BACKGROUND AND DESIGN GAP

A. Corpus Construction and Quality Selection

C4, RefinedWeb, FineWeb, and Dolma document how extraction, language filtering, content rules, and deduplication shape large language-model corpora [1]–[4]. DataComp-LM makes curation recipes comparable under controlled training protocols [5]. Learned approaches then treat quality as a selection objective. QuRating learns four scalar ratings from pairwise model judgments and uses them for sampling and curriculum construction, while D4 combines deduplication with embedding-based diversification [12], [13]. Empirical studies further show that a one-dimensional quality classifier is not sufficient to predict filtering effects across domains and need not measure an intrinsic notion of document quality [6], [7].

Our framework complements learned rankers by governing how their evidence may affect membership. It targets settings in which every change needs a closed explanation and distinct false-removal risks remain noncompensatory: a Quality score cannot repair an invalid input or justify eliminating the only surviving member of a support family.

B. Curation Systems

Data-Juicer provides composable operators, recipe exploration, tracing, and evaluation feedback [8]. NeMo Curator supplies Ray-based pipelines, heuristic and model filters,

and exact, fuzzy, and semantic deduplication across multiple modalities [10]. Oasis combines interactive rule filters, a neural filter, adaptive deduplication, and holistic corpus assessment [9]. These systems establish operator-level extensibility and scalable execution. Our focus is orthogonal: a cross-policy membership-authority contract that can sit above such mechanisms and can be implemented with their custom stages.

The contract is evaluated through its decision boundaries rather than a throughput comparison: retrieval supplies candidates, an external teacher supplies bounded Quality evidence, Coverage supplies only a veto, and external evaluation receives a materialized corpus without a feedback path. Table I turns these boundaries into executable counterfactuals.

C. Deduplication for Code

Deduplication can reduce memorization and evaluation leakage [14]. The Stack applies responsible source-code dataset construction and near-deduplication [15]. MinHash and LSH efficiently retrieve high-overlap candidates [16], but overlap is especially ambiguous for code. Two files can share imports and structure while implementing different behavior, and pairwise near relations need not be transitive. We therefore use similarity only to decide which pairs receive a deterministic relation test. It never supplies membership authority by itself.

D. Two Failure Patterns

Two composition failures motivate the contract. Functions differing only in `return x >= 0` versus `return x > 0` may collide in many LSH bands although neither represents the other; retrieval must therefore open a comparison rather than authorize deletion. Conversely, every member of a partial-fragment group may lack enough local Q1 evidence without being known incorrect. Materializing those local non-selections can silently erase the group, so Coverage alone may require a representative. The first case over-authorizes similarity and the second over-interprets missing positive evidence. Table I makes both failure modes executable rather than leaving them as pipeline-order advice.

III. EVIDENCE-BOUND MEMBERSHIP MODEL

A. Evidence, Policy, and Authority

Let D be an audited input corpus and $U(D)$ the deterministic training units created from it. A profile m materializes $C_m \subseteq U(D)$ together with a trace T_m . For a unit x , an evidence producer emits an observation $e = (x, h_x, p, v, r)$ containing the unit identity, text hash, producer, version, and result. An observation is descriptive until an authorized policy maps it to a typed state transition.

The model recognizes four non-compensatory roles:

- 1) **Validity** may quarantine a closed input-contract failure.
- 2) **Redundancy** may propose removal of a nonrepresentative only with an authorized relation witness and surviving representative.
- 3) **Quality** may positively select a unit under independent Q1–Q4 evidence; insufficient positive evidence is

TABLE I

EXECUTABLE AUTHORITY COUNTERFACTUALS. CANDIDATE-EDGE COUNTS ARE NOT UNIQUE WOULD-BE DELETIONS; THEY QUANTIFY DECISIONS THAT THE CONTRACT REFUSES TO INFER FROM RETRIEVAL ALONE.

Shortcut	Unconstrained membership outcome	Evidence-bound restriction	Frozen audit result
Retrieval as deletion	Similarity candidates become removable duplicate-family members	Typed relation witness and final surviving representative	Normal 2,515 $\rightarrow$ 2; Hard 2,873 $\rightarrow$ 3
Local selection only	Threshold-not-met units disappear even when a support group reaches zero survivors	Coverage has veto-only restore authority followed by full recheck	99 Normal and 429 Hard chunks restored
Stale evidence reuse	A changed text, provider, policy, or threshold inherits an earlier decision	Every consumed artifact is identity- and hash-bound	All exercised identity mutations rejected
Evaluation feedback	A favorable or adverse score retunes membership after observing outcomes	Evaluator fields are absent from selector inputs; corpus hashes freeze first	60 cells observed with zero feedback path

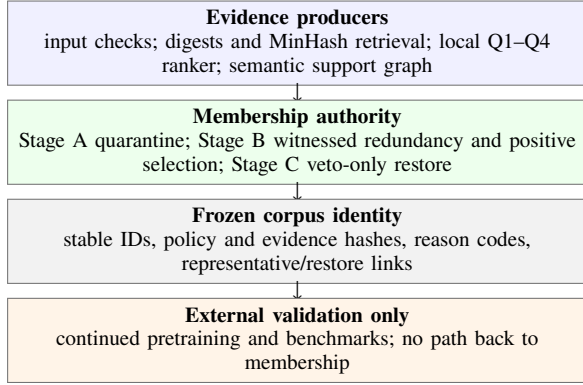


Fig. 1. Evidence can reach corpus membership only through the authority shown for its role. The external experiment observes frozen outputs and has no selector input path.

recorded as *not selected*, not as proof that the unit is useless.

- 4) **Coverage** may veto the combined proposals and require explicit rematerialization; it cannot create a new exclusion.

Figure 1 shows the separation. The boundary is asymmetric: uncertain Redundancy evidence retains, whereas uncertain Quality evidence does not count as a positive pass. Both behaviors are explicit policy choices, not implicit consequences of one global score.

B. Required Invariants

For each membership-changing proposal $p(x)$, the trace must contain a valid policy artifact π , evidence e , and text identity h_x :

$$p(x) \Rightarrow \text{valid}(\pi, e, h_x) \wedge \text{authorized}(\pi, e, x, m). \quad (1)$$

For Redundancy, $p(x)$ additionally names a representative $s(x)$ and witness $w(x, s)$, with $s(x)$ surviving final materialization. Missing evidence cannot be replaced by a run-local heuristic. A changed tokenizer, normalization, provider, model artifact, threshold, or schema invalidates inherited evidence.

Let V be Stage B-eligible units and P_m the reason-coded Stage B non-selection set. The provisional corpus is $\tilde{C}_m = V \setminus P_m$. For every active support group g , Coverage requires

either a survivor or an authorized zero-survivor explanation a_g :

$$I_g(\tilde{C}_m) = \mathbf{1}[g \cap \tilde{C}_m \neq \emptyset \vee a_g = 1]. \quad (2)$$

If any $I_g = 0$, Stage C emits required-retain IDs R_m , writes $C_m = \tilde{C}_m \cup R_m$, and reruns the complete invariant set. No weighted average allows one intact group to compensate for another group's extinction.

C. Executable Contract Invariants

An accepted materialization exposes five checks already enforced by the runtime contract. Every membership change must emit a complete *reason-coded trace* naming its policy, observable evidence, action, and text identity. Every Redundancy exclusion must pass *representative survival*. Coverage runs the declared *zero-survivor check* and emits explicit required-retain IDs when support would otherwise disappear. The *forbidden-input check* rejects selector access to downstream outcomes. Finally, *identity-bound replay* requires identical bound artifacts for resume and rejects inherited evidence after a text, provider, policy, or threshold identity changes.

Proposition 1 (contract-checked materialization). Assume deterministic unitization and policy execution, collision-resistant content identities, and complete trace emission. Any transaction accepted by the final recheck satisfies the five invariants above; in particular, retrieval-only deletion, representative-free Redundancy removal, unexplained support extinction, evaluation-driven reselection, and stale-evidence replay cannot be accepted.

Proof sketch. Induct over the ordered transition log. Stage A can emit only a closed Validity quarantine. Every Stage B removal is admitted by the role-specific policy and records the evidence and current text hash; Redundancy additionally records a survivor checked against final membership. Stage C forms only $\tilde{C}_m \cup R_m$ and reruns Eq. (2), so it cannot originate an exclusion. The external evaluator is downstream of the frozen corpus hash and has no selector field. Finally, any changed bound identity aborts resume before a transition is consumed. These cases exhaust the membership-changing transitions in the contract. The proposition is a property of accepted executions under the stated assumptions, not a claim that an implementation is free of arbitrary software defects.

The distinction between *rejected* and *not selected* remains part of the trace. Validity and witnessed Redundancy carry closed negative evidence; a Quality threshold-not-met state says only that the profile lacks sufficient positive Q1–Q4 evidence. No role may borrow another role’s signal to compensate for a failed precondition.

D. Forbidden Feedback

The curation runtime is intentionally unable to read benchmark outcomes, pretraining loss, NLL, a utility estimate, source reputation, source tier, target domain shares, a retention fraction, or a maximum token budget. These values can describe or evaluate a frozen output, but cannot alter that output. This is stronger than promising not to tune on a benchmark: the forbidden inputs are absent from the active policy configuration and reported as unread in materialization artifacts.

IV. CURATION PIPELINE

A. Stage A: Closed Validity

The input adapter preserves declared metadata without inferring missing rights or provenance. Stage A normalizes only payload-preserving representation and quarantines observable contract failures: absent text, a declared text-contract violation, unrecoverable corruption, or acquisition failure. Short code, partial files, unfamiliar formats, markup, multilingual text, and parser failure remain valid by default. Rights, PII, poisoning, and source-admission review are separate audit concerns; they are not silently converted into a text-quality score.

Released records are chunked deterministically into units of at most 6,000 characters. Every chunk retains its parent record ID and normalized text hash. Stage B first removes normalized exact duplicates, before candidate retrieval or model-assisted Quality is considered.

B. Stage B: Witnessed Redundancy

Candidate retrieval combines exact text and formatting digests, exact containment-window digests, and MinHash LSH. The active retrieval settings are 5-token shingles, a 32-value signature, eight bands, and a 24-token retrieval minimum. Normal admits bounded-near comparisons from 96 tokens with at most two changed tokens and 1%; Hard uses 64 tokens, four changes, and 2%. Units of at most 32 tokens are exact-only.

Retrieval only schedules comparisons. An exclusion proposal requires an authorized typed witness: exact text, formatting equivalence, bounded near substitution, token-preserving prose reflow, exact token containment, or, in Hard, a declared equivalence verifier. Numeric, operator, negation, answer label, API-signature, code-identifier/keyword, and named-entity changes trigger substantive-difference protection. Near edges remain pairwise and are not transitively closed. A payload superset is preferred as representative, with a stable ID tie break.

This split between retrieval and decision is essential to the paper’s claim. The frozen run evaluated thousands of candidate pairs, yet every authorized redundancy exclusion was justified

by exact token containment rather than MinHash similarity alone (Section VI-C).

C. Stage B: Structural Non-Payload Rules

Four closed text rules detect an explicit generated-and-do-not-edit artifact, a license-comment-only chunk, an empty HTML shell, or a chunk containing only a fixed vocabulary of web controls. Each requires positive in-text evidence. A generated source label alone, ordinary documentation, code with a license header, prose about cookies, and HTML with visible or executable content are protected negative cases. Candidate span transformations are disabled in the frozen Normal and Hard runs; consequently, all selected chunks are exact substrings of their parent records.

D. Stage B: Distilled Vector Quality

Quality is an evidence vector, not an intrinsic scalar:

$$Q(x) = (Q_1, Q_2, Q_3, Q_4), \quad Q_i \in \{P, F, A\}, \quad (3)$$

where P , F , and A denote pass, fail, and abstain. The heads ask whether the unit has (Q1) locally observable correctness evidence, (Q2) semantic coherence, (Q3) substantive payload, and (Q4) at least one learnable relation. A head fails only on its closed reason codes. Missing external knowledge, undeclared execution assumptions, missing context, specialized notation, low confidence, and out-of-distribution input abstain.

GPT-5.6 Luna supplies offline Q1–Q4 labels through a versioned Batch configuration. It does not process the full corpus and its output alone may neither select nor exclude a chunk. The fitting pool combines 1,024 corpus chunks with 576 policy-targeted synthetic enrichment fixtures (144 per head), giving each head 1,168 labeled observations before its fit/calibration split. A UID- and text-hash-disjoint protected set contains 800 corpus chunks. The local four-head ranker uses Qwen3-Embedding-0.6B representations, the same encoder and heads for both profiles, minimum decision confidence 0.8, and OOD cosine-similarity threshold 0.3822.

Let $P(x)$ be the confident in-distribution heads predicting pass and $F_m(x)$ the qualified fails at operating point m . The selection rules are

$$\text{Select}_N(x) = \mathbf{1}[|P(x)| \geq 1 \wedge |F_N(x)| = 0], \quad (4)$$

$$\text{Select}_H(x) = \mathbf{1}[|P(x)| \geq 2 \wedge |F_H(x)| = 0]. \quad (5)$$

Abstention supplies no positive selection evidence. Therefore a chunk that does not meet the pass count is recorded as threshold-not-met, not as a qualified failure or a claim of universal uselessness. Stage C may later restore it only through a typed Coverage veto.

E. Stage C: Veto-Only Coverage

Coverage evaluates the combined Stage B result over nonexclusive views: redundancy families, deterministic content routes, language/script and format labels, residual links, and semantic support strata. The frozen semantic graph uses

Qwen3-Embedding-0.6B as its primary provider and BGE-M3 as an independent audit view [22], [23]. Mutual eight-neighbor relations shared by both views define stable local support; provider-specific neighborhoods remain uncertainty strata. Embedding similarity never authorizes an exclusion.

When a provisional output leaves a group unexplained and empty, Coverage chooses a deterministic representative, records why it is required, restores it, and performs a second complete check. Hard restoration candidates are bounded by final Normal membership, enforcing

$$C_{\text{Hard}} \subseteq C_{\text{Normal}}. \quad (6)$$

Coverage does not maximize a diversity score, impose a quota, or infer that a rare group is useful. Its claim is narrower: membership loss must be accounted for, and an unexplained extinction blocks materialization.

F. Audit Trace

Each boundary writes stable unit and parent IDs, original and evidence hashes, policy/provider versions, discrete decisions, reason codes, and any representative or required-retain link. The top-level report aggregates token and chunk impact by reason and checks representative survival, subset invariants, residual identity, and complete Coverage recheck. Composition is reported only between Stage B eligibility and the Stage C output so the unit and denominator remain comparable.

Materialization is executed as a transaction:

- 1) validate source, tokenizer, policy, model, and schema identities;
- 2) release or quarantine records under Stage A and deterministically create chunk identities;
- 3) retrieve Redundancy candidates, evaluate typed witnesses, apply closed non-payload rules, and run the version-bound local Q1–Q4 ranker;
- 4) write the complete provisional membership set and all reason traces;
- 5) evaluate every Coverage view, emit required-retain IDs, explicitly rematerialize, and rerun the complete invariant set; and
- 6) hash the final corpus and only then expose it to the external training protocol.

An interrupted run may resume only when every bound identity matches. A stale or corrupt cache reduces usable evidence or aborts the transaction; it cannot activate a looser fallback. This recovery rule makes reproducibility a membership property rather than a best-effort logging feature.

V. IMPLEMENTATION

A. Content-Addressed Execution

The implementation separates policy specification, evidence artifacts, and materialization. JSON contracts bind the source manifest, tokenizer, chunking, model and provider identities, operating points, and output locations. A runtime bridge rejects a resume if a bound identity or content hash changes. The final

TABLE II
AUTHORITY AND FAIL-CLOSED BEHAVIOR IN THE IMPLEMENTED PIPELINE.

Role	Authorized action	Fallback
Validity	Quarantine a closed contract failure	Release on ambiguity
Redund.	Propose witnessed removal; link survivor	Retain pair
Quality	Select on pass count and no qualified fail	No positive pass
Coverage	Veto and explicitly restore support	Never excludes
External eval.	Measure a frozen corpus/model arm	No feedback path

TABLE III
VERSIONED IMPLEMENTATION SETTINGS THAT AFFECT MEMBERSHIP.

Component	Setting
Identity	Stable UID, parent UID, normalized text/evidence SHA-256
Chunking	Format preserving; maximum 6,000 characters
Retrieval	Exact/format/containment digests; 5-token shingles; 32-value MinHash; eight LSH bands
Redundancy	Exact-only through 32 tokens; Normal ≥ 96 tokens, ≤ 2 changes/1%; Hard ≥ 64 , $\leq 4/2\%$
Non-payload	Four closed in-text rules; no active span transformation
Quality	Qwen3-Embedding-0.6B; confidence ≥ 0.8 ; OOD cosine ≥ 0.3822 ; Normal ≥ 1 pass, Hard ≥ 2
Qualified fail	Q1/Q3 fail off; Q2 Normal/Hard 0.4236/0.2032; Q4 0.8150/0.3602
Coverage	Qwen3 embedding primary, BGE-M3 audit view; mutual eight-neighbor support graph; veto only
Selector inputs	Text and authorized evidence only; no utility, loss, benchmark, source, quota, or token budget

Normal and Hard configurations use the same active policy families; only versioned Redundancy bounds and the required Quality pass count differ.

Table III lists settings that materially affect membership. Luna is absent from the full-corpus execution row because it is an offline calibration oracle. Likewise, Qwen3-4B-Base appears only in external training, never as a selector feature.

B. Policy Qualification Boundary

The distilled ranker is a candidate policy component. Its 1,024-dimensional encoder representation, four heads, OOD boundary, and protected-set threshold checks are inspectable. Q1 and Q3 have no promoted fail threshold; Q2 and Q4 alone use protected-set-verified fail thresholds. The evaluated corpus records no qualified Quality failure, so its profile difference comes from one-pass versus two-pass positive selection. We treat this as a fixed Code-corpus operating point, not as a universal quality classifier.

Accordingly, the paper reports two evidence levels: executable conformance of the membership transaction and a bounded Code-domain materialization and training study. Production release and cross-domain promotion remain outside the present evidence; they are not prerequisites for testing the proposed decision contract.

C. Conformance Evidence

Fixtures exercise positive, false-positive, adversarial, and metamorphic paths. Corpus reports then verify typed Quality decisions, final representatives, veto-only Stage C, complete recheck, and the Hard-subset-of-Normal invariant; source, leakage, and packed-input reports bind hashes and exact token, block, and step counts. These checks establish executable conformance, not corpus-wide semantic accuracy or downstream benefit.

The frozen semantic artifact records 3,843 stable strata, 2,830 uncertainty strata, and 23,636 pairwise entries under two provider identities. Normal and Hard bind the same graph hash and every required-retain UID. These are the support views Stage C consumed, not an estimate of their semantic precision.

VI. EVALUATION

A. Corpus and Exclusion Boundary

The frozen Raw arm is a mixed Python-code corpus, not a purely raw Stack sample. It contains 4,228 records and 4,723,925 Qwen3-4B stream tokens (67.64%) from `bigcode/the-stack-dedup`, plus 662 records and 2,260,513 tokens (32.36%) from version-pinned snapshots of eight established open-source repositories. The final total is 4,890 records and 6,984,438 stream tokens including one EOS per record.

Before curation, an exclusion audit compared 4,902 candidates with frozen snapshots of HumanEval+, MBPP+, LiveCodeBench Code Generation Lite, BigCodeBench Complete, CRUXEval-I/O, and DS-1000. It excluded 12 records using exact normalized segments or shared 16-token shingles. This establishes only the declared exact/shingle boundary; it does not prove absence of semantic, translated, shorter, or base-model contamination.

Calibration and protected Quality samples are UID- and text-hash-disjoint from each other but drawn from the same 8,024-chunk corpus. Corpus-wide materialization then uses the frozen local ranker. The external teacher is not queried, and all configuration reports state that benchmark, utility, and token-budget inputs were unread.

B. Natural-Budget Continued Pretraining

We continue Qwen3-4B-Base [21] on audited Raw, Normal, and Hard. Each arm uses its complete natural token stream; there is no equal-token resampling, target retention fraction, or maximum-token cap. Seeds are 101, 202, and 303. Training uses sequence length 2,048, micro-batch size 1, gradient accumulation 8, learning rate 5×10^{-5} , weight decay 0.1, gradient clipping 1.0, and QLoRA rank 32, alpha 64, and dropout 0.05 over all linear target modules. Adapter training uses 4-bit NF4 weights, double quantization, and bfloat16 compute. Only the final incomplete sequence and accumulation group are dropped. The model and tokenizer are bound to revision 906bfd4b4dc7f14ee4320094d8b41684abff8539. Optimization uses AdamW with betas (0.9, 0.999), epsilon 10^{-8} , a constant learning rate, and no scheduler or warmup.

All reported scores use deterministic pass@1 generation. Prompts are tokenized without added special tokens; decoding is greedy and stops at EOS or the suite-specific maximum: 512 new tokens for EvalPlus, 1,024 for BigCodeBench and DS-1000, and 256 for CRUXEval. EvalPlus, BigCodeBench, and DS-1000 use the same code-fence/terminal-marker trimming rule; CRUXEval uses its canonical direct prompts and upstream task-specific postprocessing. Scoring binds EvalPlus 0.3.1, BigCodeBench v0.1.4 and its public Complete evaluator, CRUXEval commit 190faf1, and DS-1000 commit b39aab7. The frozen protocol, packed-input report, generated task IDs, scorer outputs, and all 60 cell hashes are retained in the accompanying manifest package.

The primary reasoning suite is BigCodeBench Complete, CRUXEval-I, CRUXEval-O, and DS-1000. HumanEval+ and MBPP+ are mandatory secondary short-function diagnostics [17]–[20]. The hierarchy was timestamp-frozen after all HumanEval+ and partial MBPP+ results and Base-only CRUXEval observations, but before any non-Base result on the primary suite. This is a post-EvalPlus, pre-primary-results analysis amendment, not a preregistered endpoint. All arms, seeds, and six benchmarks remain reportable. The primary summary is the unweighted macro of the four benchmark percentages after seed aggregation. No arm or seed may be omitted because of its outcome, and the two secondary diagnostics remain visible even when they disagree with the primary summary.

C. Operational Validation

Table IV records the materialized corpora used by every downstream run. Raw unit count is a record count, while Normal and Hard are chunk counts; only token columns support direct size comparison. Normal retains 87.70% of Raw stream tokens and Hard retains 72.05%. Packed tensors contain 2,984 and 2,456 sequences respectively, compared with 3,408 for Raw.

Table VI exposes the decision path. Stage A released 4,889 of 4,890 records, and Stage B produced 8,024 eligible chunks after one exact duplicate rejection. The closed structural rules removed 39 explicit generated artifacts and one license-comment-only chunk in each profile. Normal evaluated 2,515 redundancy candidate pairs and authorized two containment removals; Hard evaluated 2,873 and authorized three. All had stable representatives, and no removal was authorized solely by MinHash, a semantic neighbor, or a transitive near-duplicate component.

The Quality stage is a positive-selection gate. Normal selected 7,048 chunks before Coverage and recorded 934 as threshold-not-met. Hard selected 5,430 and recorded 2,551 as threshold-not-met. In both profiles only 25 chunks passed all four heads, and almost every chunk contained at least one abstention. The run recorded no qualified fail. Those facts prevent an incorrect interpretation of the nonselected set as a measured set of bad programs.

Stage C restored 99 Normal and 429 Hard chunks. Both outputs then passed the complete recheck, no transformation was applied, and Hard remained an exact UID subset of

TABLE IV

FROZEN NATURAL-BUDGET MATERIALIZATION. UNIT TYPES DIFFER BETWEEN RAW (RECORDS) AND CURATED ARMS (CHUNKS); STREAM AND PACKED TOKENS USE THE SAME QWEN3-4B TOKENIZER.

Arm	Units	Stream tokens	Packed tokens	Stream retention	Blocks	Steps
Raw audited	4,890 records	6,984,438	6,979,584	100.00%	3,408	426
Normal	7,147 chunks	6,125,213	6,111,232	87.70%	2,984	373
Hard	5,859 chunks	5,032,400	5,029,888	72.05%	2,456	307

TABLE V

ARTIFACT-AUDITED BENCHMARK SCORES (%). TRAINED-ARM COLUMNS REPORT SEEDS 101, 202, AND 303 SEPARATELY. THE FINAL ROW REPORTS THE THREE-SEED PRIMARY-MACRO MEAN $\pm$ SAMPLE STANDARD DEVIATION; NO BEST-SEED SELECTION IS USED.

Benchmark	Base	Raw			Normal			Hard		
		101	202	303	101	202	303	101	202	303
HumanEval+	31.10	30.49	28.66	27.44	32.93	32.93	29.27	29.88	32.93	29.27
MBPP+	58.47	45.50	50.00	51.32	54.76	52.65	53.70	54.76	54.76	53.97
BigCodeBench	39.91	40.35	41.84	40.79	40.18	38.95	39.39	39.12	40.09	40.35
CRUXEval-I	3.50	6.38	7.63	7.50	8.75	5.50	4.75	5.63	6.25	6.63
CRUXEval-O	2.13	2.88	2.00	1.38	5.63	3.25	5.00	5.38	5.25	4.13
DS-1000	28.00	33.10	33.60	32.90	31.60	32.20	31.90	31.10	33.00	32.70
Primary macro	18.38	20.68	21.27	20.64	21.54	19.97	20.26	20.31	21.15	20.95
Mean $\pm$ SD	–	20.86 $\pm$ 0.35			20.59 $\pm$ 0.83			20.80 $\pm$ 0.44		

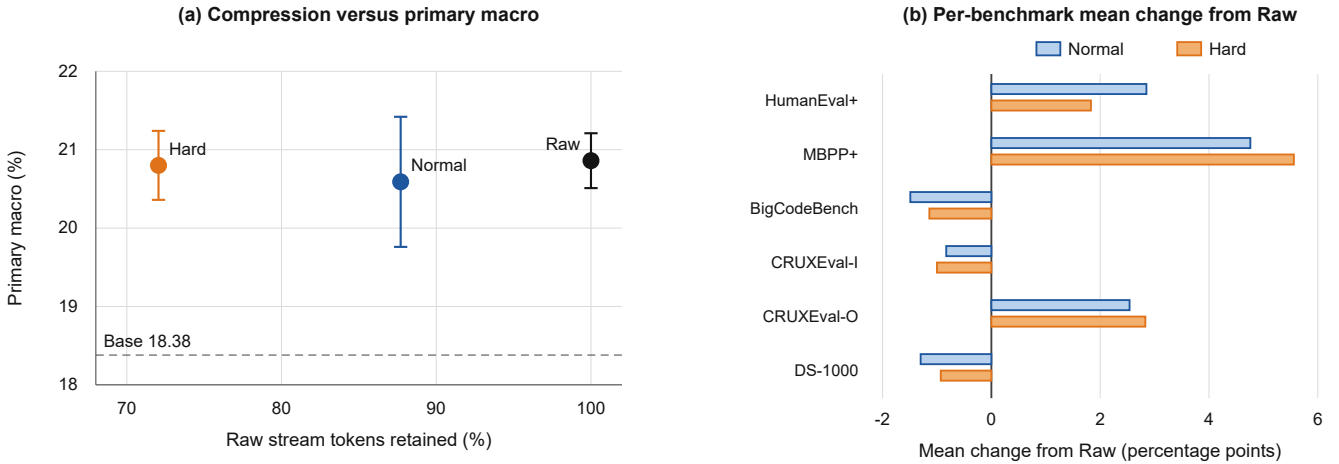


Fig. 2. Compression and capability trade-offs. Panel (a) reports three-seed means with sample-standard-deviation error bars; Base is a no-update reference, not a zero-token training arm. Panel (b) reports descriptive mean differences. Neither panel is a significance or non-inferiority test.

Normal. The restore counts directly demonstrate that the Coverage path was active rather than decorative; they do not by themselves establish that every restored semantic group improves downstream performance.

The counterfactuals expose why the operators alone are not the contribution. Of 2,515 Normal and 2,873 Hard retrieved Redundancy pairs, only two and three respectively satisfied an authorized relation witness. Thus, 2,513 and 2,870 candidate edges remained evidence for comparison rather than silently becoming membership decisions; these are edge counts, not unique would-be deletions. Without the Stage C veto, 99 Normal and 429 Hard required-retain units would be absent from provisional membership and the declared support in-

variant would fail. Finally, the identity-mutation checks reject inherited evidence rather than resuming under a changed text, provider, policy, or threshold identity.

D. Source and Composition Effects

Source identity was not exposed to the selector, yet retention is source-correlated. Normal retains 85.64% of Stack tokens and 92.00% of reference-pool tokens; Hard retains 69.40% and 77.59%, respectively. Reference records are also longer (median 6,283 versus 2,039 characters), so they account for 25.61% of Stage B chunks despite being 13.54% of Raw records. We report record, chunk, and token denominators separately and do not infer a causal source-quality effect.

TABLE VI
MECHANISM AUDIT FOR THE FROZEN NORMAL AND HARD OUTPUTS.

Measure	Normal	Hard
Stage B-eligible chunks	8,024	8,024
Structural non-payload exclusions	40	40
Redundancy candidate pairs	2,515	2,873
Witnessed redundancy exclusions	2	3
MinHash-only authorized exclusions	0	0
Quality threshold-not-met	934	2,551
Quality-selected before Coverage	7,048	5,430
Coverage required-retain IDs	99	429
Final chunks	7,147	5,859
Unexplained extinction after recheck	0	0
Active text transformations	0	0

Using the comparable chunk view from Stage B eligibility to the Stage C output, the exclusive primary route distribution moves only modestly. Code-artifact share is 49.90% at eligibility, 49.84% in Normal, and 49.56% in Hard; mixed code/prose is 43.42%, 44.33%, and 44.46%; general prose is 5.61%, 4.98%, and 5.23%. These deterministic route labels explain composition and are neither semantic ground truth nor target proportions.

E. Downstream Evaluation and Interpretation

Table V reports Base once and preserves the 101/202/303 seed order for every trained arm. The unit of comparison is the across-seed arm summary, not an individual run. The timestamp-frozen primary statistic is the unweighted macro over BigCodeBench Complete, CRUXEval-I, CRUXEval-O, and DS-1000 after seed aggregation. HumanEval+ and MBPP+ are mandatory secondary diagnostics and cannot be dropped in response to an unfavorable result.

Figure 2 makes the compression and capability trade-offs explicit. All 60 model-benchmark cells passed artifact-level provenance checks covering 42,820 task judgments: generated task identifiers matched scored identifiers, each task had one judgment, and every reported score was recomputed from the task-level outcomes. The primary reasoning macro is 18.38 for Base, 20.86 for Raw, 20.59 for Normal, and 20.80 for Hard. Thus, Hard retains 72.05% of Raw stream tokens while finishing 0.06 percentage points below Raw on the primary macro; Normal retains 87.70% while finishing 0.27 points below Raw. All three updated arms exceed Base on this summary. Paired Hard-Raw seed differences are -0.37 , -0.12 , and $+0.31$ points, exposing variability hidden by the small mean difference. Secondary diagnostics show a capability redistribution: curation recovers much of Raw’s MBPP+ regression and improves CRUXEval-O, while BigCodeBench and DS-1000 remain below the Raw mean. Because benchmark-specific non-inferiority margins and a one-sided paired rule were not frozen, the three-seed means and sample standard deviations are descriptive rather than a formal non-inferiority result.

The matrix answers a policy-qualification question outside the framework. Comparable performance with fewer natural tokens qualifies a bounded compression result; gains and re-

TABLE VII
EVIDENCE AVAILABLE NOW AND THE CORRESPONDING CLAIM BOUNDARY.

Evidence	Supported interpretation
Typed traces and forbidden-input audit	Implemented separation of evidence and membership authority
2,515/2,873 comparisons; 2/3 witnessed removals	Retrieval was active; no MinHash-only removal in this corpus
99/429 explicit restores and passing recheck	Coverage veto affected materialization and preserved declared support invariants
87.70%/72.05% natural-token retention	Exact corpus effect, not intrinsic quality or equal-budget efficiency
Timestamped three-seed protocol	All arms and tasks are reportable; outcomes cannot alter membership

gressions identify the capability trade-off of that frozen profile. The precommitted boundary keeps these observations from retroactively changing its policy, threshold, corpus identity, or membership trace.

F. Claim Boundary

The mechanism evidence establishes separate decision authority, satisfied trace and Coverage invariants, and exact changes in corpus membership and token volume. The downstream matrix then qualifies the frozen profiles for the declared corpus, model, recipe, and benchmark hierarchy. It does not isolate a causal membership effect or compare execution-engine superiority. Table VII keeps each interpretation adjacent to its evidence.

VII. DISCUSSION

A. Contribution Relative to Existing Curation Systems

QuRating and related learned rankers strengthen evidence production, while NeMo Curator and Data-Juicer provide broad operator libraries and scalable execution. Our contribution occupies a different layer: it specifies when an operator output is allowed to become a membership decision. A ranker can be replaced, a retrieval index can be distributed, and a support graph can be rebuilt without granting any of them undeclared authority. The retrieval-as-authority, no-Coverage, and identity-mutation counterfactuals make this distinction executable rather than terminological. The framework can therefore be implemented on top of an existing execution engine instead of competing with it on throughput. Its Big Data role is a control-plane contract: distributed operators may produce evidence independently, while content identities and typed transitions preserve the provenance of large numbers of otherwise irreversible local decisions. The present study validates that contract but does not benchmark distributed control-plane overhead.

B. What the External Experiment Can Decide

The mechanism audit and the training experiment serve different evidentiary roles. Invariant checks establish that the declared transaction was executed; they do not show that the

resulting corpus helps a language model. Conversely, a benchmark gain can qualify one frozen policy but cannot validate an arbitrary Quality signal or prove that every excluded unit was harmful. Natural-budget training is essential here because it preserves the operational consequence of curation: Normal and Hard expose the model to fewer tokens. The result matrix must therefore be read jointly with compression, seed variability, and the short-function diagnostics rather than reduced to one favorable cell.

The framework also permits an informative negative result. If a profile regresses across the fixed primary analysis, its membership contract remains auditable but that profile is not promoted as an effective policy. This separation prevents downstream outcomes from being used to rewrite the selector after the fact and keeps engineering conformance distinct from empirical qualification.

The observed pattern is a capability redistribution rather than a uniform gain. Raw has the highest primary macro but substantially regresses on MBPP+ relative to Base. Normal and Hard recover most of that short-function loss and more than double Base and Raw on CRUXEval-O, while remaining below Raw on BigCodeBench and DS-1000. Because source retention and sequence boundaries are confounded with membership, these contrasts motivate targeted ablations; they do not identify which individual Quality or Coverage decision caused a score.

VIII. LIMITATIONS

Four evidence boundaries guide interpretation and follow-up work. First, evaluation does not fully isolate membership selection. Raw is a designed mixture whose Stack component was already deduplicated, limiting observable redundancy removal. Reference-pool retention is higher despite source identity being hidden, so length, style, model familiarity, and payload remain confounded. Raw also appends EOS at whole-record boundaries, whereas Normal and Hard use chunk boundaries. A Chunked-All arm is therefore the highest-priority control for separating boundary effects from membership effects.

Second, statistical and semantic validation remain corpus-bound. The analysis hierarchy was amended after HumanEval+ and partial MBPP+ were visible, although it was frozen before any non-Base primary-suite result. Quality fitting uses 576 policy-targeted synthetic fixtures; its 1,024 corpus labels and 800 protected chunks originate from the same corpus. Stored observations make the proprietary teacher replayable but not independently reproducible. Likewise, semantic Coverage providers define candidate support mechanisms rather than semantic ground truth. Independent cross-corpus, multilingual, and multidomain false-veto audits would test transfer of both components.

Third, two artifact qualifications affect release rather than the reported transition trace. One valid 112-token Django record was quarantined by a pre-fix acquisition-failure rule; the corrected runtime removes that rule, but the frozen models use the earlier outputs. In addition, all 662 reference records have external repository-level license and commit documentation

while their frozen JSONL lacks complete record-level rights metadata and hashes for those fields. Rights are absent from selector inputs, but a self-contained public corpus release requires hash-bound metadata repair.

Finally, the experiment is smaller than production pretraining: one 6.98M-token Code corpus, one 4B base model, and QLoRA continued pretraining. The study validates the control-plane contract but does not measure distributed runtime, memory, or trace-storage overhead, and contamination analysis covers the declared exact/shingle boundary. The downstream result consequently qualifies this corpus, model, training recipe, and scoring protocol; broader effectiveness requires independently frozen domains and controls.

IX. CONCLUSION

Evidence-bound curation makes corpus membership auditable by assigning distinct authority to closed Validity, witnessed Redundancy, positive vector Quality, and veto-only Coverage. The Code instantiation demonstrates that contract in operation: candidate retrieval never became similarity-only deletion, every Redundancy exclusion named a surviving representative, Coverage restored 99 Normal and 429 Hard chunks, and both materializations passed complete rechecks without target size or downstream utility. All 60 benchmark cells also passed artifact-level provenance checks covering 42,820 task judgments.

Across three seeds, Hard retained 72.05% of Raw tokens and reached a 20.80 primary reasoning macro, compared with 20.86 for Raw; Normal retained 87.70% and reached 20.59. The mixed task-level pattern characterizes compression with near-retained reasoning performance and capability redistribution rather than uniform improvement. These results qualify the frozen Code policies while preserving the broader contribution: a domain-neutral authority layer in which external evaluation may assess a materialized corpus but cannot redefine how membership decisions are made.

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